

Group Project Proposal (Max 2 Pages)

Below is the outline for your research-oriented group project. The objective of this project is to enhance your group collaboration skills, conduct a comprehensive literature review, identify scientifically intriguing research gaps, and formulate and evaluate well-defined research questions. Your one-page proposal should address the sections 1-6 below. References are not included in the page limit.

Proposal Document Outline

Title: *Clear, descriptive, and specific (≤ 15 words)*

Team: *Names, emails*

1) Motivation

- **What problem matters and why now?**

Briefly situate the problem in deep learning / reinforcement learning (DL/ RL) or the application domain (e.g., healthcare imaging, robotics control).

- **Practical or scientific relevance:**

State the outcome that a successful project would enable (e.g., faster training, safer exploration, better generalization).

Tip: Avoid generic claims—include a concrete user, system, or metric that would benefit (e.g., “reducing sample complexity in sparse-reward navigation by $\geq 20\%$ ”).

2) Gap in the Literature

- **What do prior works do?**

Summarize the closest approaches (algorithms, architectures, evaluation protocols).

- **What’s missing?**

Identify a limitation (e.g., restricted to tabular RL, heavy compute

assumptions, weak out-of-distribution generalization, lack of reproducible baselines).

- **Your angle:**

State the precise gap your project targets (e.g., “no controlled study of entropy-regularized REINFORCE vs off-policy DQN under matched compute on stochastic grid worlds with curriculum learning”).

Tip: Name 2–3 anchor papers you will build upon (to be fully cited in the final report). Keep this section focused on **comparability** (shared metrics/datasets) to anchor your methodology.

3) Research Question(s)

- Pose **one primary** and (optionally) **one secondary** question that are **testable**.
 - *Example:* “Does adding a value-function baseline to REINFORCE improve sample efficiency relative to DQN under sparse rewards?”

Tip: Avoid “prove X is better.” Prefer “under which conditions does X outperform Y?”

4) Experimental Methodology

- **Setting & Data/Env:**
Specify datasets or environments (e.g., CIFAR-10, Gymnasium GridWorld, Gymnax/Brax variants), and why they are appropriate.
- **Models & Algorithms:**
Outline the architectures and algorithms to compare (e.g., DQN, REINFORCE with/without baseline, PPO as a sanity check if time permits).
- **Controls & Baselines:**
Include a random/heuristic baseline and at least one strong prior baseline.
- **Hyperparameter Search:**
Define the search space (e.g., learning rates $\{10^{-4}, 10^{-3}, 10^{-2}\}$; discount factors $\{0.9, 0.99\}$) and selection protocol (validation performance; fixed budget).

- **Seeds & Statistics:**
Plan for **10 independent seeds** per condition; report mean $\pm$ standard error; include individual seed traces.
- **Metrics & Plots:**
RL: average return vs episodes; sample efficiency (episodes to reach threshold); policy heatmaps/arrows.
DL: accuracy, calibration (ECE), training time/throughput.
- **Compute & Reproducibility:**
Detail hardware (Morningstar, personal GPU, Colab); pin framework versions; publish a deterministic run script; provide a GitHub repo with README, `environment.yml/requirements.txt`.
- **Risk & Mitigation:**
Name one risk (e.g., unstable training) and a fallback (e.g., reduce network size, gradient clipping, simpler env).

Tip: Prefer **matched compute** across methods (same wall-clock budget or steps) and **identical evaluation protocols**.

5) Expected Contributions

- A reproducible **comparison** clarifying when/why method A beats method B.
- An **implementation** (JAX-first, with clean baselines) others can reuse.
- An **insight** (ablation or sensitivity) about architecture/hyperparameters or environment design.

6) Work distribution

- Outline what each member of the group will contribute (eg. Student A and B will lead literature review, All group members will develop research questions and experimental method, Student A and D will implement and evaluate research question 1, etc.)

Formatting & Submission

- **Length:** strictly **1 page** (excluding references; single column; ~600–800 words).
- **File:** PDF, 11-pt font, 1-inch margins.
- **Due:** February 24, 2026
- **Submit:** Brightspace Group Project Proposal

Evaluation (Proposal, Max 2 pages — 100 pts)

- **Motivation & Clarity (15 pts):** clear, concrete problem and stakes.
- **Literature Gap (20 pts):** focused, comparable prior work; well-posed gap.
- **Research Question(s) (10 pts):** testable, scoped, aligned with gap.
- **Methodology (35 pts):** sound design (baselines, seeds, hyperparams, metrics, reproducibility, compute plan).
- **Expected Contributions & Timeline (10 pts):** feasible, valuable, planned.
- **Writing Quality (10 pts):** concise, structured, jargon controlled.